

Overview of Semantic segmentation

Subject: Semantic segmentation

1.

Fast marching methods The fast marching method has been used in image segmentation, and this model has been improved (permitting both positive and negative propagation speeds) in an approach called the generalized fast marching method.

2.

This above equation is used to determine whether a point in the image is an isolated point based on the response magnitude $|R(x, y)|$ and a threshold value T . If the response magnitude is greater than or equal to the threshold, the function returns 1, indicating the presence of an isolated point; otherwise, it returns 0. This helps in the effective detection and segmentation of isolated points in the image.

3.

Semi-automatic segmentation In one kind of segmentation, the user outlines the region of interest with the mouse clicks and algorithms are applied so that the path that best fits the edge of the image is shown. Techniques like SIOX, Livewire, Intelligent Scissors or IT-SNAPS are used in this kind of segmentation. In an alternative kind of semi-automatic segmentation, the algorithms return a spatial-taxon (i.e. foreground, object-group, object or object-part) selected by the user or designated via prior probabilities.

4.

Region-growing methods Region-growing methods rely mainly on the assumption that the neighboring pixels within one region have similar values. The common procedure is to compare one pixel with its neighbors. If a similarity criterion is satisfied, the pixel can be set to belong to the same cluster as one or more of its neighbors. The selection of the similarity criterion is significant and the results are influenced by noise in all instances. The method of Statistical Region Merging (SRM) starts by building the graph of pixels using 4-connectedness with edges weighted by the absolute value of the intensity difference. Initially each pixel forms a single pixel region. SRM then sorts those edges in a priority queue and decides whether or not to merge the current regions belonging to the edge pixels using a statistical predicate. One region-growing method is the seeded region growing method. This method takes a set of seeds as input along with the image. The seeds mark each of the objects to be segmented. The regions are iteratively grown by comparison of all unallocated neighboring pixels to the regions. The difference between a pixel's intensity value and the region's mean, δ , is used as a measure of similarity. The pixel with the smallest difference measured in this way is assigned to the respective region. This process continues until all pixels are assigned to a region. Because seeded region growing requires seeds as additional input, the segmentation results are dependent on the choice of seeds, and noise in the image can cause the seeds to be poorly placed. Another region-growing method is the unseeded region growing method. It is a modified algorithm that does not require explicit seeds. It starts with a single region A_1 —the pixel chosen here does not markedly influence the final segmentation. At each iteration it considers the

neighboring pixels in the same way as seeded region growing. It differs from seeded region growing in that if the minimum δ is less than a predefined threshold T then it is added to the respective region A_j . If not, then the pixel is considered different from all current regions A_i and a new region A_{n+1} is created with this pixel. One variant of this technique, proposed by Haralick and Shapiro (1985), is based on pixel intensities. The mean and scatter of the region and the intensity of the candidate pixel are used to compute a test statistic. If the test statistic is sufficiently small, the pixel is added to the region, and the region's mean and scatter are recomputed. Otherwise, the pixel is rejected, and is used to form a new region. A special region-growing method is called λ -connected segmentation (see also λ -connectedness). It is based on pixel intensities and neighborhood-linking paths. A degree of connectivity (connectedness) is calculated based on a path that is formed by pixels. For a certain value of λ , two pixels are called λ -connected if there is a path linking those two pixels and the connectedness of this path is at least λ . λ -connectedness is an equivalence relation. Split-and-merge segmentation is based on a quadtree partition of an image. It is sometimes called quadtree segmentation. This method starts at the root of the tree that represents the whole image. If it is found non-uniform (not homogeneous), then it is split into four child squares (the splitting process), and so on. If, in contrast, four child squares are homogeneous, they are merged as several connected components (the merging process). The node in the tree is a segmented node. This process continues recursively until no further splits or merges are possible. When a special data structure is involved in the implementation of the algorithm of the method, its time complexity can reach $O(n \log n)$, an optimal algorithm of the method.

5.

Markov random fields The application of Markov random fields (MRF) for images was suggested in early 1984 by Geman and Geman. Their strong mathematical foundation and ability to provide a global optimum even when defined on local features proved to be the foundation for novel research in the domain of image analysis, de-noising and segmentation. MRFs are completely characterized by their prior probability distributions, marginal probability distributions, cliques, smoothing constraint as well as criterion for updating values. The criterion for image segmentation using MRFs is restated as finding the labelling scheme which has maximum probability for a given set of features. The broad categories of image segmentation using MRFs are supervised and unsupervised segmentation.

6.

Graph partitioning methods Graph partitioning methods are an effective tools for image segmentation since they model the impact of pixel neighborhoods on a given cluster of pixels or pixel, under the assumption of homogeneity in images. In these methods, the image is modeled as a weighted, undirected graph. Usually a pixel or a group of pixels are associated with nodes and edge weights define the (dis)similarity between the neighborhood pixels. The graph (image) is then partitioned according to a criterion designed to model "good" clusters. Each partition of the nodes (pixels) output from these algorithms are considered an object segment in the image; see Segmentation-based object categorization. Some popular algorithms of this category are normalized cuts, random walker, minimum cut, isoperimetric partitioning, minimum spanning tree-based segmentation, and segmentation-based object categorization.